Greedy play can require more colors in the coloring game

P. Graham
https://orcid.org/0009-0007-8493-1505

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Abstract

In the coloring game, Alice and Bob alternately color vertices properly from a fixed palette, with Alice trying to color the whole graph. The game chromatic number is the least palette size for which she can win. In the greedy game the players choose vertices, each receiving the least available positive integer; Alice minimizes and Bob maximizes the number of colors used. Havet and Zhu asked in June 2011 whether the resulting game Grundy number is always at most the game chromatic number. We give an explicit finite simple graph G for which $\chi_g(G) \le 5 < \Gamma_g(G)$, answering the first part of their question negatively. The proof consists of independently checked finite strategy certificates for the two unrestricted Alice-first games. It does not establish arbitrarily large separation between the parameters.

Computer-assisted; not yet reviewed by experts.

The proofs, the search code, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 The question and result

The ordinary coloring game is played on a finite simple graph with a palette of k colors. Alice and Bob alternately give an uncolored vertex a color different from those of its colored neighbors. Alice wins if the whole graph is colored; otherwise Bob wins. The least k for which Alice has a winning strategy is the *game chromatic number* $\chi_g(G)$. In the greedy game, a chosen vertex receives the least positive integer absent from its colored neighbors. Alice minimizes and Bob maximizes the largest color used. The value is the *game Grundy number* $\Gamma_g(G)$. Alice moves first in both games, either player may choose any uncolored vertex, and neither player may pass.

Theorem 1. *The graph G defined in Section 3 is a finite simple graph with* 11 *vertices and* 27 *edges satisfying*

$$\chi_g(G) \le 5 < \Gamma_g(G).$$

Both inequalities concern unrestricted vertex selection with Alice moving first, alternating turns, and no passes.

Havet and Zhu introduced the game Grundy number in a report first posted on June 15, 2011. The first sentence of Problem 5 in version 1 reads:

> "Is it true that $\Gamma_g(G) \le \chi_g(G)$ for all graph G?"

The problem then asks separately whether the game Grundy number can be arbitrarily large compared with the game chromatic number [1, p. 6]. Theorem 1 answers the first part negatively. It

does not answer the second part, and makes no assertion about the exact values of either parameter for G or about the smallest possible counterexample.

Hollom restated the inequality as Question 3.4 and explained an obstacle to transferring an ordinary winning strategy to greedy play [2]. His ordered-game example concerns a prescribed sequence of vertex locations. It does not itself give an unrestricted counterexample. The new contribution here is an explicit graph together with an ordinary winning strategy for Alice and a greedy strategy for Bob that survives every legal choice of vertex by Alice. This universal coverage of Alice's choices is the essential additional obligation beyond an ordered example; it is supplied by the finite certificates.

There is also a distinct question in the reverse direction. Krawczyk and Walczak proved that the game chromatic number is not bounded by a function of the ordinary, static Grundy number [3, Theorem 7]. The static Grundy number is the maximum number of colors used by greedy coloring over all vertex orders, so it is at least $\Gamma_g(G)$. Their theorem therefore also rules out bounding $\chi_g(G)$ by a function of $\Gamma_g(G)$. This earlier result concerns the reverse boundedness question, rather than the inequality answered here. The present proof uses the definitions and question of Havet and Zhu and the distinction emphasized by Hollom; it does not use the construction of Krawczyk and Walczak.

Section 2 describes the proof and the finite games used in the certificates. Section 3 specifies the graph. Section 4 gives the certificate argument, and Section 5 describes the reproducible checks.

2 Games and proof overview

The proof compares two games on the same graph with the same five colors; it does not compare separate graph families or rely on changing the palette of an ordinary strategy.

For example, if a vertex has colored neighbors using color 1 and no other color, an ordinary move with five colors may give it any of $2, 3, 4, 5$. The greedy move must give it color 2. Although greedy moves form a subset of ordinary moves at a fixed position, both players are subject to this restriction. An ordinary winning strategy for Alice may ask her to make a move that the greedy rule forbids.

Write $V(G)$ for the vertex set and $N_G(v)$ for the neighbors of a vertex v. A palette-k position is a function

$$s : V(G) \longrightarrow \{0, 1, \ldots, k\},$$

where 0 denotes an uncolored vertex and adjacent nonzero values must differ. For an uncolored vertex v, define its available colors by

$$A_k(s, v) = \{1, \ldots, k\} \setminus \{s(u) : u \in N_G(v),\ s(u) \neq 0\}.$$

An ordinary legal move chooses any uncolored v and any $c \in A_k(s, v)$. A greedy legal move in the palette-k game chooses any uncolored v with $A_k(s, v) \neq \varnothing$ and uses $c = \min A_k(s, v)$. In each case the new position changes only $s(v)$, from 0 to c.

The certificates stop a palette-k game as soon as either all vertices are colored, an Alice win, or an uncolored vertex has no available color, a Bob win. Such a vertex is called *blocked*. An ordinary game may instead be described as continuing until no legal move remains. These conventions have the same winner: once a vertex is blocked, its neighbors' colors never disappear, and a full coloring is impossible.

The unrestricted greedy game has no fixed palette. A selected vertex always receives the least positive integer absent from its colored neighbors, so a move is possible until every vertex is colored.

Its used colors form an initial interval of the positive integers: a first use of color c requires colored neighbors of every smaller positive color. Consequently its largest color and its number of colors agree.

The proof has two parts. The ordinary certificate selects a legal move for Alice and covers every legal move by Bob. It forces a full proper coloring from the five-color palette. The greedy certificate selects a legal move for Bob and covers every greedy move by Alice. It forces a blocked vertex in the palette-five game. Since the five colors on that vertex's neighbors persist, unrestricted greedy play must eventually use a color at least six.

Each certificate is a directed graph of literal labeled positions. Different histories reaching the same position share its continuation. There is no quotient by graph automorphisms or by color permutations. Every arc colors one additional vertex, so correctness reduces to a finite induction. The argument establishes two specific strategies; it requires neither a search over all graphs nor an exhaustive solution of every position in either game.

3 The graph

The following list specifies the example independently of its drawing. Its vertices are $0, 1, \ldots, 10$. In the right column of Table 1, each entry v in row u denotes the edge $\{u, v\}$; only neighbors with larger labels are listed. Thus each edge appears exactly once, and there are no additional edges.

u	Neighbors greater than u	u	Neighbors greater than u
0	$3, 5, 7, 8$	6	$9, 10$
1	$4, 6, 8, 9, 10$	7	9
2	$5, 6, 7, 9, 10$	8	10
3	7	9	10
4	$6, 8, 9, 10$	10	$\varnothing$
5	$7, 8, 10$		

Table 1: The complete edge list of G, grouped by smaller endpoint.

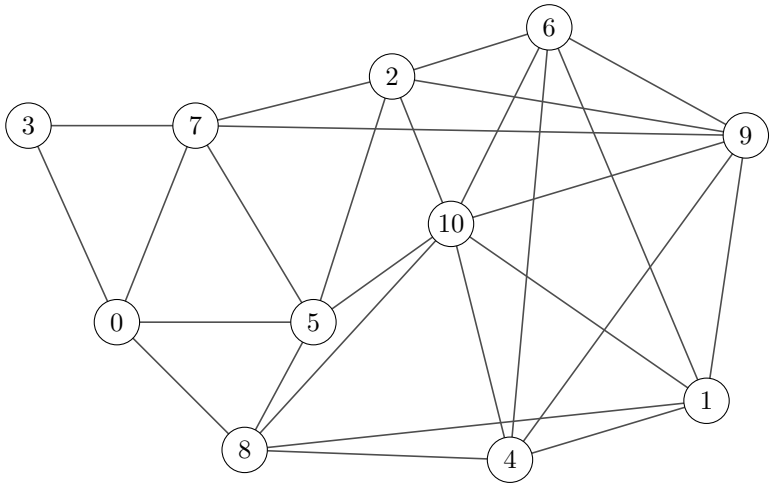


Figure 1: The graph G. Crossings in the drawing are not vertices. The drawing is generated from the edge list in `data/graph.json`.

The machine-readable graph is `data/graph.json`. Both strategy certificates contain the same vertex set and edge list, and the public entry point checks their equality to this graph before checking either strategy. It also binds both certificates to palette size five, Alice-first play, and unrestricted vertex selection. The ordinary certificate's first move colors vertex 5 with color 1; subsequent choices are given by its positions and arcs, rather than by a short case-based strategy in the text.

4 Why the certificates prove the theorem

This section gives the finite induction behind the checks and explains why the greedy certificate implies a lower bound in the unrestricted game.

4.1 Local strategy obligations

A strategy certificate declares a game, a starting player, a palette size, a graph and a claimed winner. Its nodes are proper partial colorings. The root is the all-zero position. A node lists moves (v, c), each leading to the position obtained by assigning color c to the previously uncolored vertex v. Alice moves when the number of colored vertices is even; Bob moves when it is odd.

At every reachable node the checker recomputes all legal moves directly from the graph and the partial coloring. A terminal node lists no moves, and its recomputed winner must be the declared winner. At a nonterminal node belonging to the declared winner, the certificate lists exactly one legal move. At a nonterminal node belonging to the other player, it lists exactly all legal moves, without duplication. Every listed successor must be present, and all nodes must be reachable from the root.

Lemma 1. *Consider either palette-k game defined in Section 2, with Alice first and with stopping at a blocked vertex. If a finite certificate satisfies the local strategy obligations above, its declared winner has a winning strategy from the all-zero position.*

Proof. For each certificate position s, induct on the number of its uncolored vertices. A terminal position has the declared winner by the terminal check. At a nonterminal position where that winner moves, the selected legal move leads to a position with one fewer uncolored vertex, which is winning by induction. At a position where the opponent moves, all legal moves are included, and every successor is winning by induction. This proves the claim at every reachable position, including the root. The strict increase in the number of colored vertices along each arc also rules out cycles. □

4.2 From a blocked vertex to a sixth greedy color

The lower bound uses a threshold certificate, so its connection with the unrestricted greedy game needs a separate argument.

Lemma 2. *Let H be a finite simple graph. Suppose Bob has a winning strategy in the Alice-first greedy palette-k game of Section 2, with unrestricted vertex selection, alternating turns, no passes, and stopping when a vertex is blocked. Then Bob can force a color greater than k in the unrestricted greedy game on H. In particular, $\Gamma_g(H) > k$.*

Proof. Until a blocked vertex occurs, every uncolored vertex has some available color in $\{1, \dots, k\}$. Its least available positive integer is therefore the same in the unrestricted game and in the palette-k game. Bob follows his palette-k strategy during this common part of play. Every choice of vertex by Alice is a legal choice in the threshold game and is covered by that strategy.

Bob's winning strategy reaches a blocked vertex v before a full coloring. At that point every color in $\{1, \ldots, k\}$ occurs on a colored neighbor of v. These colors persist. The unrestricted game continues until all vertices are colored, since each turn colors one previously uncolored vertex and no player may pass. When v is colored, its greedy color is consequently greater than k. Thus Bob forces the required lower bound, regardless of subsequent moves. □

Proof of Theorem 1. Table 1 specifies a simple graph of the stated order and size. The two checks described in Section 5 accept an ordinary certificate with declared winner Alice and a greedy certificate with palette $k = 5$ and declared winner Bob for this graph and the stated conventions.

By Lemma 1, Alice has a winning ordinary strategy with five colors. Hence $\chi_g(G) \leq 5$ directly from the definition as the least winning palette size. No assertion that ordinary winning palettes are monotone is needed. The same lemma gives Bob a winning strategy in the greedy palette-five game. Lemma 2 then gives $\Gamma_g(G) > 5$. □

5 Finite verification and reproducibility

The bundle contains the graph, the two finite strategies and two separately written certificate checkers. Checking requires Python 3 and its standard library, with no graph generator, search executable or external mathematical package. From the bundle's top-level directory, the command is

```
python3 -I checks/verify.py
```

The same command checks both certificates and performs the corruption tests. Detailed format and checker documentation is in `checks/README.md`.

5.1 Data and checker independence

The files `data/ordinary-k5.json.gz` and `data/greedy-k5.json.gz` are gzip-compressed JSON objects. Besides the game conventions and graph, each contains a map from comma-separated color vectors to lists of moves. The coordinate order is the vertex order $0, \ldots, 10$; 0 is the uncolored marker. All states and moves are labeled literally. A strategy can discard unneeded choices at its winner's turns, while retaining every legal choice at the opponent's turns. The resulting sizes are reported in Table 2; these are certificate sizes, not full game-tree sizes.

Game	Declared winner	Nodes	Arcs	Terminal nodes
Ordinary, $k = 5$	Alice	108 669	216 995	2 800
Greedy, $k = 5$	Bob	5 594	6 730	2 010

Table 2: The two finite strategy certificates.

The programs `checks/check.py` and `checks/check_independent.py` implement the local obligations of Section 4 separately. Neither imports the search program or the other checker. They reconstruct adjacency, properness, available colors, the player to move, legal moves and terminal outcomes from each certificate. In particular, terminal winners are recomputed; they are not trusted labels attached to leaves. The entry point also checks the graph and game bindings against the public statement's five-color obligations. Thus acceptance of a different graph, a different starting player or an ordered variant cannot substitute for the two obligations used in the theorem.

5.2 Corruption tests and trusted components

For each certificate the public command makes seven semantic changes: it omits an opponent reply, reverses the declared winner, inserts an illegal Alice move, adds an edge, changes the starting player, treats the root as a terminal node, and adds a move at a terminal node. Each altered certificate is submitted directly to both checkers, and both must reject it. The command therefore requires 28 rejection outcomes in addition to accepting the two unchanged certificates. These controls exercise game semantics rather than file-integrity checks.

The proof relies on the correctness of the two Python implementations, the Python interpreter and its JSON and gzip libraries, and the written finite-induction and blocked-vertex arguments above. The certificate checks use exact integer and finite-set operations. The graph drawing is generated from the graph data, and its edges are checked against that data; the mathematical definition of G remains the edge list. There is no proof-assistant formalization. The search that found the graph is unnecessary for checking the theorem, and no exhaustive graph search or claim of minimality is part of the result.

References

[1] Frédéric Havet and Xuding Zhu, *The game Grundy number of graphs*, Research Report RR-7646, INRIA, version 1, first posted June 15, 2011. Problem 5, p. 6. https://inria.hal.science/inria-00600738v1/document.

[2] Lawrence Hollom, *On monotonicity in Maker-Breaker graph colouring games*, arXiv:2308.03528, version 2. First posted August 7, 2023; revised November 8, 2024. Question 3.4. https://arxiv.org/abs/2308.03528v2.

[3] Tomasz Krawczyk and Bartosz Walczak, *Asymmetric coloring games on incomparability graphs*, arXiv:1503.04748, first posted March 16, 2015. Theorem 7. https://arxiv.org/abs/1503.04748.